Fiscal Year:	FY 2008	**Task Last Updated:**	FY 01/25/2009
PI Name:	Rolston, Steven L Ph.D.		
Project Title:	Quantum Transport in Optical Lattices		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20742-4111	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	08/01/2003	**End Date:**	07/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received extension to 7/30/2008 per PI (1/2009) NOTE: End date changed to 9/30/2006 from original end date of 11/30/2008		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNC04GA07G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This work will study the coherent quantum transport of atoms trapped in periodic optical potentials. It will investigate quantum transport in classically chaotic systems, to probe the quantum-classical transition. It will study the effects of spatial and temporal noise added in a controlled fashion to the periodic potential. • Measure quantum transport in quantum pinball systems • Measure coherence of atoms that have tunneled large distances • Observe Anderson localization in optical potentials with added spatial noise • Study the effects of (un)correlated noise on quantum transport • Observe dynamic localization in time-modulated potentials • Evaluate 3D experiments suitable for microgravity		

	The transport of electrons in solid-state systems is at the heart of almost all technologically relevant properties. A system of atoms trapped in optical lattices offers an ideal model system in which to study transport properties, especially when quantum in nature. The defect-free and completely understood potential combined with unprecedented control over the system will allow for signifcant improvements in our understanding of such issues. This work resides at the boundary of atomic and solid-state physics. We can gain insight and benefit both areas.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will aid in the understanding of the effects of impurities and disorder on lattice systems, which govern the properties of many real-world materials. This is difficult to study with solid state systems due to the complications of sample preparation and consistency, but with optical lattices, we can fully characterize the system.
Task Progress:	Our system is based on using coherent atoms from a Bose-Einstein condensate loaded into optical lattices. We have devleopoed an experimental apparatus that forms BECs of rubidium atoms with up to 100,000 atoms in a condensate. This is accomplished with a millimeter scale magnetic trap in an UHV chamber. We have taken two approaches to adding disorder in or optical lattice systems. We have applied two extra perturbing lattices to a 1D optical lattice. These extra lattices were chosen with lattice constants incommensurate with the primary lattice. Although strictly speaking this is not disorder, for finite systems such as ours (the atoms occupy about 70 lattice sites) an incommensurate lattice geometry is equivalent to random disorder. The second method for applying disorder is through the use of laser speckle. For this to be useful requires the speckle correlation length be of the order of the lattice constant. We have develeoped and characterized an optical system that produces speckle with a correlation length of less than 0.9 microns, which should be sufficient for experimentation. We have studied the dynamics of adiabaticity in 1D optical lattices with the addition of one or two incommensurate perturbing lattices. By measuring momentum distributions, we were bale to show that even a very weak perturbing lattice greatly modified the requirements for adiabaticity in loading a lattice. Using band structure calculations, we showed that this is due to the formation of a number of closely spaced energy levels. Changes in the wavefunction due to the perturbers, coupled with the close eigenvalue spacing, lengthens the time requirements for adiabaticity by over a factor of 1000. the observed momentum distributions were consistent with calculations, and we also showed that these effects of disorder are washed out with atom-atom interactions, as the wavefunction localization that occurs due to disorder is removed by interactions.
Bibliography Type:	Description: (Last Updated: 01/26/2009)
Articles in Peer-reviewed Journals	Edwards EE, Beeler M, Hong T, Rolston SL. "Adiabaticity and localization in one-dimensional incommensurate lattices." Physical Review Letters 2008 Dec 31;101(26):260402. http://dx.doi.org/10.1103/PhysRevLett.101.260402 , Dec-2008